

SAATHI — Team 2

Complete Embeddings, Hybrid Retrieval, Reranking and Retrieval Evaluation Work Package

Operating rules

- Keep the existing ingestion package, versioning package, current AI/backend, and the ~1.25 GB RAGA model. Do not delete or retrain existing assets before benchmarking them.
- Use the existing 600+ MB government-source dataset as the controlled starting corpus. Every source must retain provenance; do not silently add unverified web data.
- Work in separate Git branches and reproducible Colab/Kaggle notebooks. Every experiment must record dataset/model/index version and metrics.
- Never replace a working baseline without a measured comparison.

Mission

- Build the evidence-retrieval engine that turns a question into the best BIS/government evidence. Do not generate final answers.

Step 1 — Baseline the current system

- Measure the existing FAISS retrieval on a fixed evaluation set before changing it.
- Record Recall@1, Recall@5, MRR, nDCG, latency and failure cases.
- Keep the baseline index so improvements can be compared objectively.

Step 2 — Full-corpus embeddings

- Use Team 1's canonical corpus rather than the smaller legacy catalog as the primary evidence index.
- Benchmark the existing embedding model first. Test at least one stronger alternative only if the baseline is inadequate.
- Store vector-to-chunk mapping, embedding model/version, dimension and corpus version.

Step 3 — Exact lexical retrieval

- Implement BM25 or equivalent keyword retrieval.
- Test exact standard numbers, clauses, QCO identifiers, scheme names and regulatory phrases.
- Support metadata filters for standard number, document type, date/version and source.

Step 4 — Hybrid retrieval

- Run dense and BM25 retrieval independently.
- Merge and deduplicate candidates while preserving all scores and metadata.
- Benchmark multiple fusion methods and keep the simplest method that improves recall.

Step 5 — Reranking

- Take roughly 20–50 candidates and apply a cross-encoder or other real reranker.
- Evaluate whether reranking improves Recall@5/MRR/nDCG and not merely subjective examples.
- Add timeout and fallback behavior.

Step 6 — Query processing

- Normalize common identifiers and spelling variants without changing their meaning.
- Detect standard numbers, clause references, QCOs, product terms and requested language.
- Route exact lookup queries differently from broad compliance questions when evaluation proves it useful.

Step 7 — Retrieval diagnostics

- For every failure label query-understanding, retrieval, metadata-filter, reranking or corpus failure.
- Create a retrieval debug record containing query, candidates, scores, selected chunks and expected evidence.
- Feed failure patterns back to Team 1 and Team 3 instead of blindly changing prompts.

Deliverables

- embedding/index builder
- FAISS/vector index
- BM25 index
- metadata store
- hybrid retriever
- reranker
- retrieval evaluation notebook
- retrieval failure report

Acceptance test

- Exact IS/QCO/clause queries retrieve the correct evidence.
- Broad semantic queries retrieve relevant clauses.
- Current-version filtering works.
- Retrieval metrics improve over the existing baseline or the team documents why a change was rejected.